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Q1. Program based on Classes and Objects

Score: 3.33 **TC: 2/3 passed**

CODE

```
import java.util.Scanner;

class Student{
    int rollNo;
    String name;
    double marks;

    void displayDetails(){
        System.out.print("Roll No: "+rollNo);
        System.out.print("\nName: "+name);
        System.out.print("\nMarks: "+marks);
    }

    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        Student st=new Student();
        st.rollNo=sc.nextInt();
        sc.nextLine();
        st.name=sc.nextLine();
        st.marks=sc.nextDouble();
        st.displayDetails();
    }
}
```

INPUT

```
101
John Doe
94.5
```

OUTPUT

```
Roll No: 101
Name: John Doe
Marks: 94.5
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    private int rollNo;
    private String name;
    private double marks;
    void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    void setName(String name){
        this.name=name;

    }
    void setMarks(double marks){
        this.marks=marks;

    }
    void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);

    }
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        Student st=new Student();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        st.setRollNo(rollNo);
        st.setName(name);
        st.setMarks(marks);
        st.displayDetails();
    }
}
```

INPUT

```
123456
rahul
95.6
```

OUTPUT

```
Roll No: 123456
Name: rahul
Marks: 95.6
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;

    }
    Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;

    }
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);

    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student st=new Student();
        st.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2= new Student(rollNo,name,marks);
        s2.displayDetails();
    }
}
```

INPUT

```
123456
ravi
67.5
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 123456
Name: ravi
Marks: 67.5
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    double basicSalary;
    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId=empId;
        this.empName=empName;
        this.basicSalary=basicSalary;
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empId=sc.nextInt();
        sc.nextLine();
        String empName=sc.nextLine();
        double basicSalary=sc.nextDouble();
        Salary s=new Salary();
        s.getEmployeeDetails(empId,empName,basicSalary);
        s.calculateSalary();
        s.displayDetails();
    }
}
class Salary extends Employee{
    double hra;
    double da;
    double netSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        netSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);
    }
}
```

INPUT

```
12134
ravi
20000.00
```

OUTPUT

```
Employee ID: 12134
Employee Name: ravi
Basic Salary: 20000.0
HRA: 4000.0
DA: 2000.0
Net Salary: 26000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    void getEmployeeDetails(int empId,String empName){
        this.empId=empId;
        this.empName=empName;

    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empId=sc.nextInt();
        sc.nextLine();
        String empName=sc.nextLine();
        double basicSalary=sc.nextDouble();
        GrossSalary g=new GrossSalary();
        g.getEmployeeDetails(empId,empName);
        g.getBasicSalary(basicSalary);
        g.calculateSalary();
        g.displayDetails();
    }
}
class Salary extends Employee{
    double basicSalary;
    void getBasicSalary(double basicSalary){
        this.basicSalary=basicSalary;

    }
}
class GrossSalary extends Salary{
    double hra;
    double da;
    double grossSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        grossSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+grossSalary);

    }
}
```

INPUT

```
12345
suba
500000
```

OUTPUT

```
Employee ID: 12345
Employee Name: suba
Basic Salary: 500000.0
HRA: 100000.0
DA: 50000.0
Gross Salary: 650000.0
```